CreatorAI Studio / GrowthPilot

Product & Engineering Guide — for new team members

Generated 6 September 2026 · from a full read of the codebase

> **What this document is.** A single onboarding reference: what the product is, every module's exact current features, what still needs building, and how the work is split across the team. Read sections 1–3 first, then your assigned stream in section 6.

1 · What the product is

An **AI marketing operating system for Indian small & medium businesses.** One place to build a brand, generate content and creative, run ads, capture and work leads, chat with customers, automate follow-ups, and see what's working. It is deliberately *not* a sales CRM — there is no deal pipeline.

Stack & scale

- **Next.js 16.2.12**, React 19.2, TypeScript (note: this is a modified Next.js — read `node_modules/next/dist/docs/` before touching routing/proxy/config)
- **Supabase** — auth, Postgres, Storage · **Stripe** — billing · **Anthropic Claude** — all text AI · **Replicate** — image/video/upscale/training · **HeyGen v3** — talking-avatar video
- 27 dashboard pages · 38 API routes · 11 database migrations · multi-tenant · credit-metered · row-level-security locked

2 · What "built" means here — read carefully

> **Built** = code written, TypeScript compiles, `npm run build` passes, the UI renders, the route responds without crashing.
>
> **Built does NOT mean** tested against live third-party services, tested with a real logged-in user, or verified end-to-end with real data. Almost the entire app was written against *placeholder* API keys. A large part of the remaining work is taking each "built" feature and making it *actually work* with real credentials, fixing the bugs that surface.

Two real bugs were already caught this way: a dead HeyGen status endpoint (the client was migrated to the v3 API), and a workspace-scoped Anthropic key that needs an extra header. Expect more.

3 · Current environment status (local)

Provider	State	Effect
Supabase URL + anon key	**set**	auth + database reads/writes work
Supabase **service-role key**	**placeholder**	credit deduction, public lead capture, cross-user notifications don't persist
Anthropic API key	**set but workspace-scoped**	every Claude feature returns 400 until `ANTHROPIC_WORKSPACE_ID` is added (code is already wired for it)
HeyGen API key	**set but $0 balance**	avatar render returns 402 — needs a $5 pay-as-you-go top-up
Replicate token	**placeholder**	paid image / video / upscale / textures / training disabled
Stripe secret + webhook	**placeholder**	checkout disabled
Meta / LinkedIn / Google OAuth	**placeholder**	social publishing + ads launch disabled
Twilio / Resend / WhatsApp Cloud	**placeholder**	SMS / email / WhatsApp automation actions disabled

Works right now with no further setup

- App boots, all 27 routes resolve, auth-guard active
- Signup / login (Supabase anon key is set)
- AI Image Studio via the Pollinations free fallback
- Brand wizard, manual + CSV lead capture, org/team management, analytics — the data-only features

Unlock order (highest leverage first)

1. **Supabase service-role key** → credits, public lead capture, notifications
2. **Anthropic workspace ID** → ~10 Claude features light up at once
3. **Replicate token** → video / upscale / textures / model training
4. **HeyGen balance** ($5) → avatar rendering end-to-end
5. **Stripe keys** (test mode) → billing
6. **Meta / LinkedIn / Google OAuth apps** → publishing + ads
7. **Twilio / Resend / WhatsApp Cloud** → messaging automation actions

4 · Every module — exact features today

Read directly from the source. "Add to dominate" = concrete high-impact features to build next, tuned to the Indian SMB market.

Module 1 — Auth & Foundation

Built now

- Email/password signup, login, forgot-password, OAuth callback, **passwordless OTP login**, sign-out
- Server-hydrated session store (zustand) — org, user, credits available app-wide
- Credit system: per-user balance, shared cost table (`src/lib/models.ts`), `credit_transactions` ledger (spent/granted), human-readable descriptions
- Security: RLS column-lock (migration 003) — users can only self-update name/avatar; credits/plan/Stripe-ids move only via the service-role client
- Rate limiter, `/api/health` probe, Sentry wiring

Add to dominate

- Google / WhatsApp / Truecaller social login (India is phone-first)
- 2FA/TOTP, session device list, "log out everywhere"
- Referral credits — invite a friend, both get credits (viral loop)
- Free-trial credits on signup + a "running low" usage meter
- Per-org audit log (who changed what) — needed to sell to agencies

Module 2 — Organizations / Teams / Roles

Built now

- Multi-tenant orgs; `org_members` with roles owner / admin / member / viewer
- Email invite flow, org switcher, active-org cookie
- RLS: viewers can't delete the org; only owner/admin mutate settings

Add to dominate

- Granular per-module permissions (can publish / can spend credits / can see leads), not just 4 fixed roles
- Agency client-workspaces — one agency account managing many sub-brands, each with a client-portal login
- Approval workflows — member drafts, admin approves before publish/send
- Shared asset library + per-workspace activity feed
- SSO / SAML for enterprise

Module 3 — Brand Intelligence (onboarding + AI analysis)

Built now

- 6-step brand wizard: Business Info → Audience → Brand Voice → Competitors → Goals → Platforms. Stores business type, target audience, tone of voice, offerings, USPs, pain points solved, competitors, primary goal, current platforms, monthly budget range.
- AI Business Analysis engine (Claude, structured JSON): brand summary, content pillars, recommended platforms, and four insight lists — insights, red flags, opportunities, recommendations (title + detail each). Computes real lead stats first (by source, by status, avg score by source, conversion rate) and feeds them in. Hard-constrained to never invent metrics; flags sparse data as a red flag.
- Output stored in `brand_profiles` + `ai_reports`; shown at `/ai-insights`, gated behind onboarding completion.

Add to dominate

- Auto-fill onboarding from a URL — paste your website/Instagram, AI scrapes and pre-fills the wizard (removes the biggest drop-off)
- Live competitor monitoring — posting cadence, promos, ad creative, pricing changes; weekly digest
- Brand voice from examples — upload 3 posts you like, AI extracts and locks the voice
- Category benchmarks — "your conversion rate is 2.1% vs 3.4% median for D2C skincare"
- Scheduled re-analysis with a "what changed since last month" diff

Module 4 — Brand Kit

Built now

- Colour swatches, font tags, logo upload (Supabase Storage), approved-keyword / banned-word editors
- Banned/approved words are injected into **every** AI prompt in the app — real brand-safety enforcement, not just storage

Add to dominate

- Auto-extract the kit from a logo/website (palette, fonts, logo variants)
- Logo-variation generator (light/dark/mono/favicon/social avatar) + background removal
- Brand-locked templates — every generated image/video/post auto-applies palette + fonts + logo
- Tone slider (formal ↔ playful) + reading-level control
- Compliance dictionary — legal disclaimers auto-appended for regulated niches (finance, health, edtech)

Module 5 — Campaigns & Ads

Built now

- AI ad-copy generation: 3 headlines (<40 chars), primary text, CTA label, audience/targeting paragraph, budget-guidance sentence — brand-aware, per platform + goal, honest about no live ad data
- Saves as draft `campaigns` rows

- Ads launch: creates a **PAUSED** campaign shell on Meta Ads (objective-mapped) or Google Ads (daily budget + PAUSED search campaign). Never auto-enabled — deliberate safety property.
- Optimisation (budget reallocation, CTR, segmentation) — gated, needs live ad-account data

Add to dominate

- Full campaign build — ad sets, targeting, creative, keywords — then one-click activate
- Creative generation in-loop — AI writes copy *and* generates the in-brand image/video for the ad
- Live performance sync — spend / CTR / CPL / ROAS pulled back, unified across Meta + Google
- AI optimisation that acts (with approval) — pause losers, shift budget, refresh fatigued creative
- WhatsApp Click-to-Chat ads as a first-class campaign type (huge in India)
- UTM builder + attribution: ad spend → leads → revenue

Module 6 — Content Studio

Built now

- AI content generation: one variation per selected platform, each with caption + hashtags + CTA. Platform-aware rules (IG/FB long + 8–15 hashtags; LinkedIn professional + 2–4; Twitter/X ≤280 + 1–3). Brand-aware, banned-word enforced.
- Save drafts to `content_posts`; schedule via `scheduled_at`
- Calendar view, publishing hub, quick-create entry
- Real publishing to Facebook Page, Instagram (2-step container/publish), LinkedIn (text-only), YouTube (resumable video upload)

Add to dominate

- **Multi-language generation** — Hindi, Tamil, Telugu, Marathi, Bengali, Hinglish (a wedge on its own)
- Calendar autopilot — AI fills a month from your content pillars + the Indian festival/sale-day calendar (Diwali, Holi, regional festivals)
- Image/video attached at generation time — full post, not just a caption
- Repurpose engine — 1 blog → 10 posts → 5 reel scripts → 1 newsletter → 1 carousel
- Carousels, Reels/Shorts, Stories, threads, first-comment scheduling; best-time-to-post from the account's own history
- Regional platforms: ShareChat, Moj, Josh, Koo

Module 7 — Landing Pages

Built now

- AI page generation: internal title, hero headline (<12 words) + subheadline, 3–4 sections (heading + body), CTA label — brand-aware, banned-word enforced
- Section editor: add / edit / reorder / delete

- Public unauthenticated render at `/lp/[slug]`
- A/B testing: Variant B (headline/subheadline/CTA), atomic view/conversion counters via a Postgres RPC, sticky per-visitor variant assigned in the proxy layer
- Public lead-capture form → leads pipeline (source = website)

Add to dominate

- Real visual templates + drag-drop builder — images, testimonials, pricing tables, FAQ accordions, countdown timers, video embed
- AI hero image / background generation in-brand
- **Custom domains + SSL** (current explicit gap; big conversion driver)
- Forms builder + **native payments** (Razorpay/UPI/Stripe) — sell a product straight from the page
- WhatsApp CTA button that opens chat with a prefilled message
- Pop-ups / exit-intent / sticky bars, SEO meta + schema, pixel + GA4 injection, heatmaps

Module 8 — Leads

Built now

- Capture: manual, CSV import (custom quote/escape-aware parser), website form, WhatsApp
- Heuristic lead scoring: base 20 + email/phone/company/high-intent-source bonuses, capped 100, with a reason string. `ai_score` / `ai_score_reason` columns exist for an AI upgrade.
- Status model new / contacted / converted / lost (extended set exists in types, UI trimmed to 4; no Kanban pipeline, deliberate)
- Tasks / follow-ups + overdue-task checker
- AI lead analysis: 2–3-sentence summary, one concrete next action, risk flag (e.g. "no contact in 5+ days despite high score") — reasons only from notes

Add to dominate

- AI scoring that actually uses the model (behaviour, engagement, firmographics)
- Lead enrichment — company size, industry, socials, email/phone validation from a name/email
- Auto-assignment + round-robin + SLA timers with escalation
- Deduplication + merge, full activity timeline (email/call/WhatsApp/page-view in one view)
- Web tracking pixel — see which leads visited which pages; high-intent alerts
- 2-way WhatsApp/email inbox on the lead record; revenue attribution back to the source campaign/post/page

Module 9 — Community / Inbox

Built now

- AI chatbot: short conversational replies, brand-context only, honest "I don't know" + human handoff, sets an escalation flag + reason on frustration/human-request/unknowns, acknowledges buying intent. Persists to `conversations` / `messages`.

- Website-widget test flow
- WhatsApp inbound webhook — parses the payload, finds/creates the lead + conversation, runs the same chatbot brain, sends the reply back (one real second channel into the same pipeline)
- Social account connections (Meta IG+FB, LinkedIn, Google/YouTube/Ads) — real OAuth2 auth-code flow, CSRF-safe state, long-lived token exchange

Add to dominate

- Deployable embeddable website chat widget (script tag) — currently only a test flow
- Unified inbox — WhatsApp + Instagram DM + Messenger + email + web chat in one thread view; AI drafts, human approves
- RAG knowledge base — upload FAQs/PDFs/catalogue; chatbot answers from real docs with citations
- **WhatsApp template messages + broadcast campaigns** (opt-in lists, 24-hour-window handling) — the killer India feature
- Product catalogue + cart + checkout inside WhatsApp; in-chat appointment booking
- Instagram comment → DM automation, keyword auto-responders, sentiment tagging, CSAT

Module 10 — AI Agents

Built now

- Marketing Agent — content/social/campaign advice; won't invent metrics
- Business Consultant — growth strategy, revenue projection only from the real converted-lead value trend (caveated), competitor analysis only from owner-entered competitors
- Sales AI Agent — per-lead qualify, objection-handling drafts, 2–3 meeting-time suggestions (can't book), ready-to-send follow-up drafts in brand tone
- All inject real org context (brand, lead stats, campaigns, latest AI report); chat history session-only

Add to dominate

- Persistent agent memory across sessions
- Agents that take actions (with approval) — "draft next week's calendar" actually creates the drafts
- Scheduled/proactive agents — a daily "here's what needs your attention" brief; anomaly alerts
- Voice mode; niche-tuned agents (D2C, real estate, edtech, clinics, restaurants)
- Analyst agent — ask questions of your data in plain language, get charts back

Module 11 — Automations

Built now

- Rule engine: triggers `lead_created`, `lead_status_changed` → ordered actions, logged to `automation_logs` with success/fail + which actions ran, run count + last-run tracked

- Actions: create follow-up task (with delay), notify assignee, send WhatsApp message (via connected number), send SMS (Twilio), send email (Resend) — all with `{name}` / `{company}` template variables
- Wired into the lead-created path + an overdue-task checker

Add to dominate

- Visual flow builder — drag nodes, branch on conditions, wait steps, A/B splits
- Many more triggers — form submit, page visit, email opened/clicked, tag added, date/anniversary, WhatsApp reply, cart abandoned, payment received, no-activity-for-N-days
- Multi-step cross-channel drip sequences with smart send-time
- Per-automation goal tracking ("12 → 4 converted, ₹80k influenced")
- Template gallery (Welcome, Re-engage cold, Post-purchase upsell, Review request, Festival offer blast)
- Webhook / Zapier / Make / n8n; quiet hours + frequency capping

Module 12 — Analytics

Built now

- Real metrics from leads / conversations / campaigns: conversion rate, average human response time (lead message → first human reply), source performance, campaign performance
- Honest — hides a tile rather than showing a fake zero when there's no data

Add to dominate

- Unified cross-channel dashboard — social reach/engagement + ad spend/ROAS + web traffic + leads + revenue in one place
- Attribution & funnel — impression → click → lead → customer → repeat, with multi-touch models
- Custom report builder + scheduled email/PDF/WhatsApp reports (agencies live on this)
- Cohort / retention, LTV, CAC, payback period
- AI narrative — "Revenue up 18%, driven by the Diwali landing page; Instagram CPL rose, shift budget"
- White-label client dashboards with shareable links

Module 13 — Notifications

Built now

- Real bell + list UI, new-lead trigger. Cross-user fan-out needs the service client wired through.

Add to dominate

- Channel preferences per user — in-app / email / WhatsApp / push / Slack
- Digest mode vs real-time, per notification type
- Smart priority — surface only what needs action; snooze; assign-from-notification
- Mobile push (PWA / app); @mentions in comments; billing/credit alerts

Module 14 — Creative Studio (7 tools)

Built now

Tool	What it does
AI Image Studio	4 providers with graceful fallback: Replicate (Flux Pro/Dev/Schnell, SDXL, Ideogram v2, Stable Cascade), OpenAI DALL·E 3 (HD), Stability Ultra + SD3 img2img, **Pollinations free fallback** (no key). Human-subject auto-detection swaps in a photojournalistic prompt + camera/lens boost + a long anatomy/skin negative-prompt. 12 style presets, negative prompts, batch count, steps/guidance, img2img with strength. Per-model credit cost 1–8.
AI Video Studio	Replicate Kling v1.6 + Luma Ray (text- and image-to-video, camera-motion param). Free fallbacks: Lightricks LTX-2 on a public HF Space (no key), and a fully-local ffmpeg Ken Burns pan/zoom on an uploaded photo (zero cost, no GPU).
AI Avatar Studio	HeyGen v3 (just migrated): create video, poll status, avatar/voice pickers. Idea → script via Claude (hook / agitation / solution / CTA, 40–90 words, brand-voice aware) with a preview-and-edit step before spending credits. 30-credit cost.
Textures & 3D	Replicate Flux-dev generates the diffuse map; a deterministic function derives height (luminance), normal (Sobel), roughness (high-freq detail), AO (local cavity). Metallic can't be derived from one image.
Upscaler	Replicate Real-ESRGAN (2×/4×, face-enhance toggle). 10–20 credits.
Model Training (LoRA)	Uploads images to a private bucket → zips → signed URL → `ostris/flux-dev-lora-trainer` on Replicate, latest version resolved dynamically, 1000 steps, trigger word, private destination model, status polled on demand.
Canvas	fabric.js in-browser editor.

Add to dominate

- One unified Studio — generate image → animate it → add captions/voiceover → post, without leaving the screen
- Brand-locked output — palette/fonts/logo auto-applied to everything
- Canva-style templated social-graphic editor with thousands of resizable templates + magic resize
- Background remove/replace, magic eraser, inpaint/outpaint, generative fill, product-photo backgrounds (huge for D2C sellers)
- Batch product-catalogue generation — CSV of products → a post/ad/graphic each
- Reels/Shorts auto-editor — script → stock/broll → captions → voiceover → music → vertical export
- Avatar overlay VFX (the deferred holographic-compositing feature); talking-avatar in 20+ Indian languages
- Real image-to-3D via a proper model; video upscaling + frame interpolation

Module 15 — Billing & Plans

Built now

- Stripe Checkout per plan; webhook grants plan + credits on success, reverts to free on cancellation; 3 configured plans (Apprentice / Artisan / Maestro)
- Per-user/creator billing model

Add to dominate

- **Razorpay / UPI / Paytm** + INR pricing + GST invoices (Stripe is painful for Indian customers)
- Usage-based credit top-ups + auto-recharge; annual plans with discount
- Free tier with watermark → conversion funnel; in-app upgrade prompts at the moment of need
- Agency/reseller billing — bill your own clients through the platform, take a margin
- Dunning, failed-payment recovery, coupons, referral credits, affiliate program

Module 16 — White Label

Built now

- Toggle in-app name + logo (agency plan + owner/admin only, server-checked); sidebar + footer read it. Custom domains out of scope.

Add to dominate

- Full white-label — custom domain, custom email sender, custom login page, remove all traces of the parent brand
- Client portal — clients log into *your* brand, see only their workspace + dashboards
- Reseller console — provision client accounts, set plans/limits, see MRR across all clients
- White-labelled PDF reports and shareable dashboards; API + widgets under the agency's brand

Module 17 — Marketplace

Built now

- Honest roadmap page linking to what already works as templates (automation templates, landing pages, content presets)

Add to dominate

- Template marketplace — creators sell automation flows, landing-page designs, content calendars, ad-copy packs; revenue share
- Service marketplace — connect SMBs with vetted freelancers/agencies
- Integrations directory — Shopify, WooCommerce, Razorpay, Tally, Zoho, Google Business Profile, Justdial, IndiaMART
- Niche starter kits — "Restaurant in a box", "Real-estate agent kit", "Coaching-institute kit"

5 · What turns this into a category leader

1. **WhatsApp Business Platform, done properly** — broadcasts, templates, catalogue, cart, checkout, chatbot, click-to-WhatsApp ads. In India this is *the* channel; owning it is the moat.
2. **True multilingual** — generate, chat, and caption in 10+ Indian languages + Hinglish, everywhere.
3. **Payments-native** — Razorpay/UPI in landing pages, WhatsApp, and invoices; sell, don't just market.
4. **Closed loop** — ad/post → lead → conversation → sale → repeat, with revenue attribution visible end-to-end.
5. **Ecommerce connectors** — Shopify / WooCommerce / IndiaMART sync products → auto-generate catalogues, ads, and posts.
6. **Google Business Profile + AI review management** — post offers, reply to reviews, track local rankings (critical for local SMBs).
7. **Mobile app / PWA** — SMB owners run their business from a phone.
8. **Agency layer** — full white-label + client portal + reseller billing is a separate, higher-value market on the same codebase.
9. **Canva-parity visual editor** so users never leave for design.
10. **Onboarding in 5 minutes** — paste a URL, get a filled brand kit + a month of content + 3 automations + a landing page, ready to review.

6 · Team work plan — 4 streams

Each person owns one stream end-to-end: validate every feature in it with real credentials, fix the bugs, and write a short test note per feature ("tested X with a real key — works" / "found bug Y — fixed in commit Z"). Stream 1 unblocks the other three, so start it first and stagger the rest.

Stream 1 — Environment & Billing Foundation

- Stand up the real Supabase project; run migrations 001–011 in order; confirm RLS.
- Add the service-role key; verify credit deduction, public lead capture, notification fan-out.
- Full auth smoke: signup → login → onboard → create org → invite a teammate.
- Wire Stripe test mode: checkout per plan, webhook grants plan + credits, cancellation reverts to free.
- Deliverable: a one-page "environment is live" note + a seeded test org others can log into.

Stream 2 — AI Text Features (Claude)

- Add `ANTHROPIC_WORKSPACE_ID`; confirm the client sends the header.
- Validate end-to-end, one by one: Business Analysis, Content Studio, Chatbot, the 3 AI Agents, Ad Copy, Landing-Page generation, Lead Analysis, Avatar script expansion.
- For each: JSON parses, the output schema matches, brand keywords/banned words are applied, error paths are graceful.
- Deliverable: per-feature test note; a shared prompt/schema reference.

Stream 3 — Creative & Media Pipelines (Replicate + HeyGen)

Work already in flight on branch `validate/generation-pipelines`.

- Order: Avatar Studio (HeyGen v3 — needs balance) → Video (Replicate + ffmpeg Ken Burns) → Image (paid Replicate path) → Upscaler → Textures → Model Training.
- For each: real render, credit deduction, `generations` row written, output URL valid, failure path handled.
- Resolve the "where does ffmpeg run in the deploy target" question early (blocks future video work).
- Deliverable: per-pipeline test note; a short decision doc on the video-jobs / worker architecture.

Stream 4 — External Integrations (Social + Messaging)

- Register OAuth apps: Meta (Instagram + Facebook, incl. ads_management), LinkedIn, Google (YouTube + Ads).
- Validate: connect flow → `social_accounts` row → real post to each platform → ads launch creates a PAUSED shell.
- WhatsApp Cloud: inbound webhook → lead + conversation + chatbot reply round-trip.
- Twilio SMS + Resend email: wire keys, test the 3 automation actions.

- Reuse `src/lib/social/oauthState.ts` for CSRF-safe state — don't re-invent it.
- Deliverable: per-integration test note; a setup guide for each OAuth app (redirect URIs, scopes).

7 · Standing rules for everyone

1. Branch per stream. PR into `main`. Never commit secrets — all `.env*` is gitignored; keep it that way.
2. This is a modified Next.js 16 — read `node_modules/next/dist/docs/` before writing routing/proxy/config code; heed deprecation notices (`middleware` → `proxy`, etc.).
3. Verify ritual: after each change run `npx tsc --noEmit`, `npx eslint <files>`, and — for anything non-trivial — `npm run build` with the dev server stopped (concurrent build + dev corrupts the `.next` cache).
4. Only one Next.js process against this `.next` dir at a time.
5. Don't trust `tsc` / `eslint` for provider correctness — actually call the real API and read the response.
6. Honest gaps, never fake results. If a provider isn't configured, show a disabled state with a one-line reason — keep that pattern.
7. Write a short test note per feature so coverage is visible at a glance.

This guide consolidates three working documents (feature inventory, module features & roadmap, team handoff) into one reference. Source of truth is always the code.